

Instrumental Variables

INTRODUCTION



Roadmap

Introductions

IV Mechanics

- Just-Identified IV

- Overidentification

- Weak vs. Many-Weak Bias

Chinese Economic Applications: Session I

Conclusions

Who Am I?

A Professor of Economics at Dongbei University of Finance and
Economics (DUFE)

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Committed to teaching rigorous, practical empirical tools to the next generation of economists

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Instrumental variables (IV) offer a principled solution: find a source of

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Featured applications: seminal papers on **Chinese economic topics** from top journals (AER, JDE, ReStud)

Session 1 Roadmap

1. **IV Mechanics**

- Just-identified IV: identification, estimation, inference
- Overidentification and 2SLS
- Weak instruments and many-IV bias

2. **Chinese Economics Applications: Session I**

- Paper 1: The China Syndrome (Autor, Dorn, Hanson 2013, AER)
- Paper 2: Roads and Growth in China (Banerjee, Duflo, Qian 2020, JPE)

3. **Conclusions and Preview of Session 2**

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In Session 2, we add a 4th: **Monotonicity** (for LATE interpretation)

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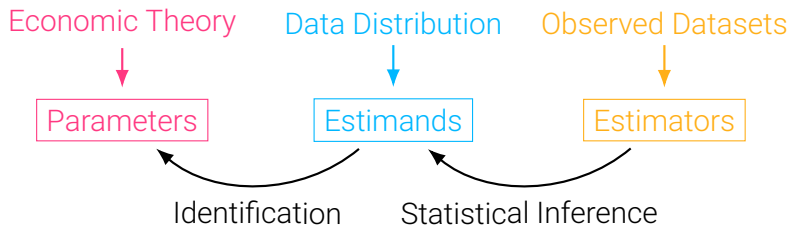
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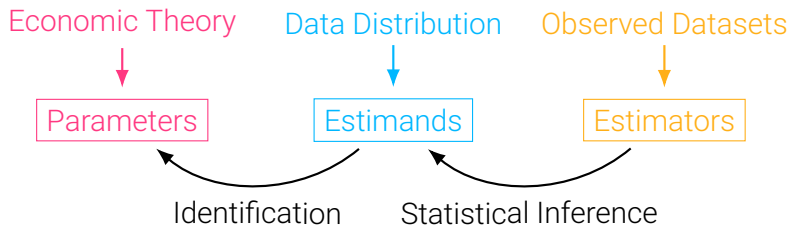
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- **Estimators** are functions of observed data (i.e. the “sample”)
 - E.g. a difference in sample means or ratio of OLS coefficients
 - Since data are random, so are estimators. Each has a distribution
 - We use knowledge of estimator distributions to learn about estimands (inference) and thus identified parameters

The Lay of the Land



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- Today we'll start with the mechanics of IV estimands, talk briefly about estimation, then spend some time talking about identification

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- We fire up Stata and `reg Y D, r`. How do we interpret the results?

Population Regression and Endogeneity

In large samples ($N \rightarrow \infty$), the OLS estimator $\hat{\beta}^{OLS}$ gets arbitrarily close to (consistently estimates) the regression estimand β^{OLS}

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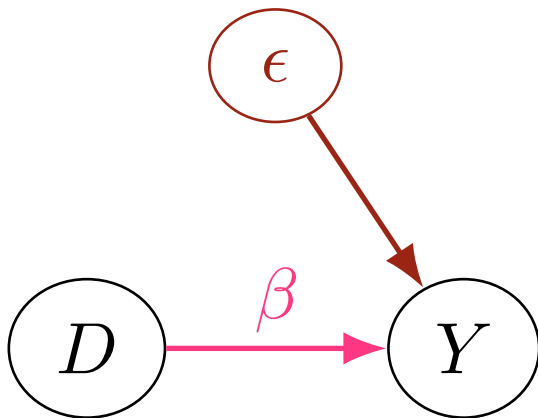
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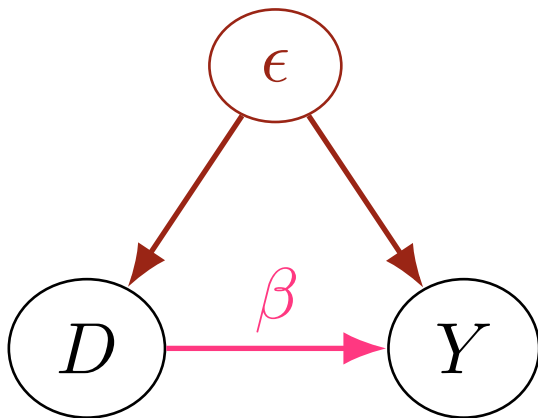
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- Selection bias: people with certain potential outcomes ε_i are more/less likely to take this course, such that $Cov(\varepsilon_i, D_i) \neq 0$

Regression “Exogeneity”



Regression “Endogeneity”



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Suppose this course was “oversubscribed” (more people wanted to attend than space allowed), so seats were determined by lottery

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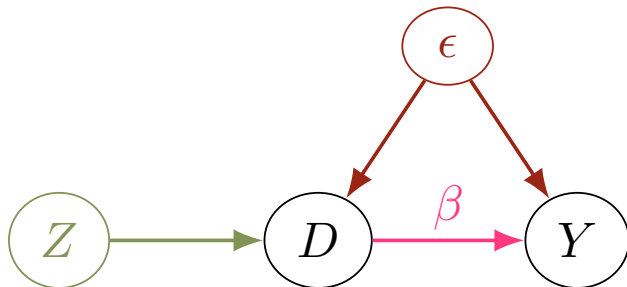
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- Or, in Stata, `ivreg2 Y (D=Z) , r`

The IV Solution



Note: no arrow connecting ϵ and Z (“as-good-as-random assignment”), and no arrow from Z to Y directly (“exclusion”). We’ll come back to both

Reduced Form and First Stage

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- But “one weird trick” links the IV estimand back to regression:

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where ρ and π come from two population regressions:

$$Y_i = \kappa + \rho Z_i + \nu_i \quad \text{The “reduced form”}$$

$$D_i = \mu + \pi Z_i + \eta_i \quad \text{The “first stage”}$$

Angrist (1990): Draft Lottery IV

Angrist famously used Vietnam-era draft eligibility as an instrument to estimate the earnings effects of military service

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- First stage $E[D_i | Z_i = 1] - E[D_i | Z_i = 0]$: effect of eligibility on the *probability* of military service (b/c D_i is binary)
- Reduced form $E[Y_i | Z_i = 1] - E[Y_i | Z_i = 0]$: effect of eligibility on adult earnings (measured in 1971, 1981...)

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IV interprets the latter causal effect in terms of the former

IV Estimates of the Effects of Military Service on the Earnings of White Men born in 1950

Earnings year	Earnings		Veteran Status		Wald Estimate of Veteran Effect
	Mean	Eligibility Effect	Mean	Eligibility Effect	
	(1)	(2)	(3)	(4)	(5)
1981	16,461	-435.8 (210.5)	.267	.159 (.040)	-2,741 (1,324)
1971	3,338	-325.9 (46.6)			-2050 (293)
1969	2,299	-2.0 (34.5)			

Note: Adapted from Table 5 in Angrist and Krueger (1999) and author tabulations. Standard errors are shown in parentheses. Earnings data are from Social Security administrative records. Figures are in nominal dollars. Veteran status data are from the Survey of Program Participation. There are about 13,500 individuals in the sample.

Adding Controls

We might only think our Z_i is exogenous controlling for some vector W_i

- Just add controls to the reduced form and first stage! $\beta^{IV} = \frac{\rho}{\pi}$ for

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- E.g. if W_i is a vector of dummies for randomization strata in an RCT, then $\tilde{Z}_i$ captures the within-strata variation in Z_i

Multiple Treatments

We might be interested in a multi-dimensional model: $Y_i = X_i'\beta + \varepsilon_i$

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- Population residuals $\tilde{Z}_i$, given some vector of controls W_i

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so long as $\text{Cov}(\tilde{Z}_i, X_i)$ is full-rank. Equivalently, $\beta^{IV} = \pi^{-1}\rho$ where:

$$Y_i = Z_i'\rho + W_i'\phi + \nu_i$$

$$X_i = \pi Z_i + W_i'\psi_i + \eta_i,$$

with estimation working just as you'd think

Multiple Instruments

What happens when $\dim(Z_i) = L > J = \dim(X_i)$? Overidentification:

$$\text{Cov}(\tilde{Z}_i, Y_i - X_i' \beta) = 0 \implies \underbrace{\text{Cov}(\tilde{Z}_i, Y_i)}_{L \times 1} = \underbrace{\text{Cov}(\tilde{Z}_i, X_i')}_{L \times J} \underbrace{\beta}_{J \times 1}$$

- We can drop any $L - J$ instruments to identify β

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$$\text{Cov}(\tilde{Z}_i, Y_i - X_i' \beta) = 0 \implies \underbrace{\text{Cov}(\tilde{Z}_i, Y_i)}_{L \times 1} = \underbrace{\text{Cov}(\tilde{Z}_i, X_i')}_{L \times J} \underbrace{\beta}_{J \times 1}$$

- We can drop any $L - J$ instruments to identify β
- More generally, we can take any full-rank linear combination

$\tilde{Z}_i^* = M \tilde{Z}_i$ for $J \times L$ matrix M such that $\text{Cov}(\tilde{Z}_i^*, X_i)$ is invertible

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This defines a *class* of IV estimands/estimators, indexed by M

Two-Stage Least Squares (2SLS)

2SLS sets $M' = \pi$: the matrix of first-stage coefficients

- This makes $\tilde{Z}_i^* = \tilde{Z}_i' \pi$ the (residualized) first-stage fitted values
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Since the first-stage from regressing X_i on this $\tilde{Z}_i^*$ is one (by construction), β^{2SLS} can be obtained in two stages:

1. Regress X_i on Z_i and W_i (first stage)
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Note: here we're talking about the 2SLS *estimand*, but the exact same logic holds for the 2SLS *estimator* (i.e. two stages of OLS)

- That said, you should never run 2SLS by hand. Let *ivreg2* do it!

Angrist-Krueger '91: The Power of Overidentification

	OLS		2SLS				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Years of education	.071 (.0004)	.067 (.0004)	.102 (.024)	.13 (.020)	.104 (.026)	.108 (.020)	.087 (.016)
<i>Covariates</i>							
9 year-of-birth dummies		✓			✓	✓	✓
50 state-of-birth dummies		✓			✓	✓	✓
<i>Instruments</i>							
dummy for QOB = 1			✓	✓	✓	✓	✓
dummy for QOB = 2				✓		✓	✓
dummy for QOB = 3				✓		✓	✓
QOB dummies interacted with year-of-birth dummies (30 instruments total)							✓

Notes: The table reports OLS and 2SLS estimates of the returns to schooling using the Angrist and Krueger (1991) 1980 census sample. This sample includes native-born men, born 1930–39, with positive earnings and nonallocated values for key variables. The sample size is 329,509. Robust standard errors are reported in parentheses. QOB denotes quarter of birth.

2SLS is a Many-Splendored Thing

There is another (I think more useful) way to understand 2SLS:
as a weighted average of just-identified IVs:

$$\begin{aligned}\beta^{2SLS} &= \left(\pi' Cov(\tilde{Z}_i, X_i') \right)^{-1} \pi' Cov(\tilde{Z}_i, Y_i) \\ &= (\pi' Var(\tilde{Z}_i) \pi)^{-1} \pi' Var(\tilde{Z}_i) \rho,\end{aligned}$$

This is the formula for a $Var(\tilde{Z}_i)$ -weighted regression of reduced-form coefficients ρ on first-stage coefficients π (with no constant)

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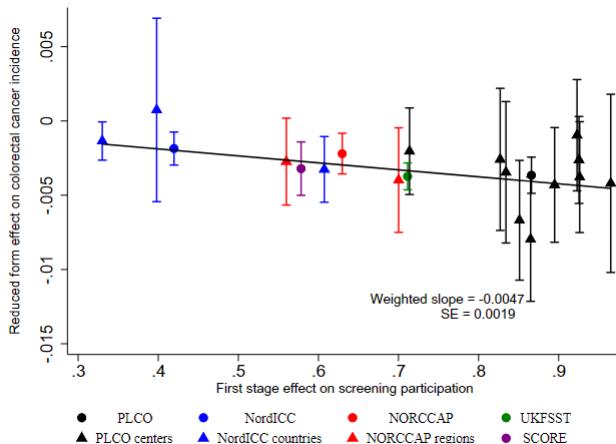
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- When $J = 1$ (one treatment), this becomes $\beta^{2SLS} = \sum_{\ell} \omega_{\ell} \beta_{\ell}^{IV}$
where $\omega_{\ell} = (\pi' Var(\tilde{Z}_i) \pi)^{-1} \pi' Var(\tilde{Z}_i)'_{\ell} \pi_{\ell}$ and $\beta_{\ell}^{IV} = \rho_{\ell} / \pi_{\ell}$
- Intuitively: 2SLS combines multiple “one-at-a-time” IVs β_{ℓ}^{IV}

Angrist-Hull '23: “Visual IV” for Cancer Screening Trials



Each dot gives a (ρ_ℓ, π_ℓ) for a trial ℓ where randomized screening offers Z_i instrument for screening participation D_i

- Slope of the weighted line-of-best fit through zero = 2SLS estimate

Overidentification Tests

Under the constant-effects causal model of $Y_i = X_i'\beta + \varepsilon_i$,
overidentification gives a way to test instrument validity

- All just-identified IVs should coincide: i.e. $\beta_\ell^{IV} = \beta$ for all ℓ
- Graphically: the R^2 from visual IV plots should $= 1$ in large samples

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Don't place too much stock in overidentification tests, however:

- They tend to have low power (b/c individual $\hat{\beta}_\ell^{IV}$ tend to be noisy)
- If they reject, it need not mean your instruments are invalid (b/c of treatment effect heterogeneity – more on this in Session 2)
- Rejection doesn't tell you which IV is invalid (they all might be!)

Weak Instruments

When running just-identified IV, people worry about instrument “strength”

- Specifically the first stage F-statistic, which tests if $\pi = 0$

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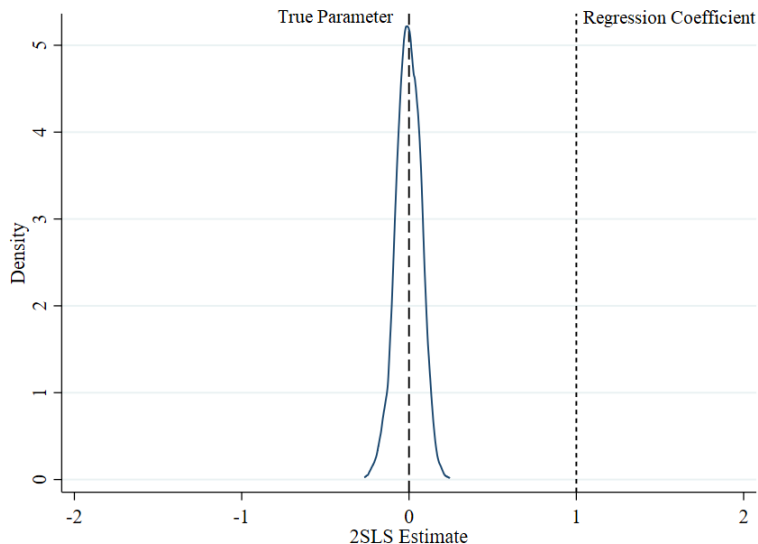
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Much has been made of this over the years, but Angrist and Kolesár (2022) show that we shouldn’t worry too much

- The SE increase tends to be large enough to “cover up” the bias, so you’re unlikely to reject the null of $\beta = 0$ spuriously

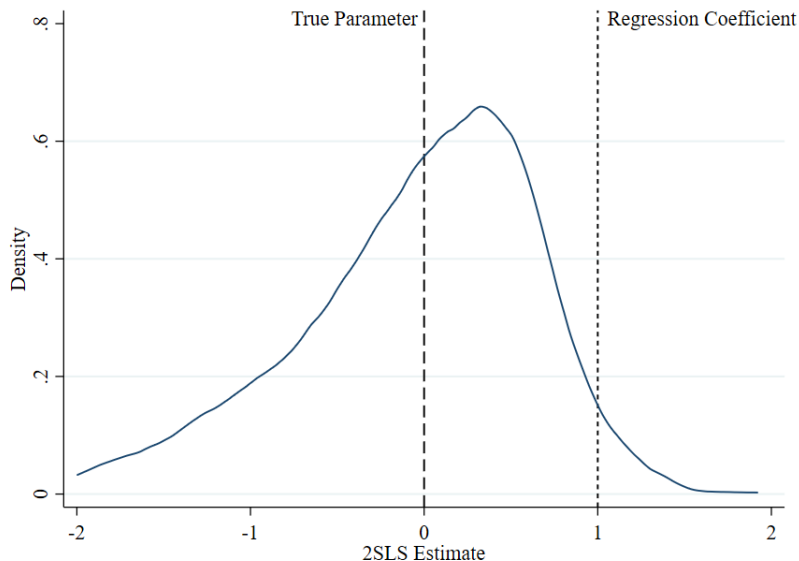
Weak Instruments: Visualized

Monte Carlo: $Y_i = 0 \cdot D_i + \varepsilon_i$, $D_i = \pi Z_i + \eta_i$: $\pi = \text{Var}(\varepsilon_i) = \text{Var}(\eta_i) = 1$



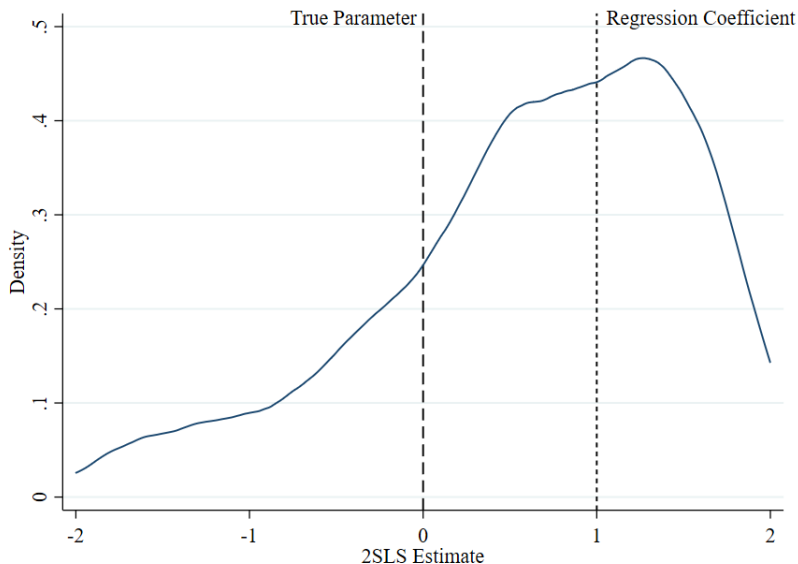
Weak Instruments: Visualized

Monte Carlo: $Y_i = 0 \cdot D_i + \varepsilon_i$, $D_i = \pi Z_i + \eta_i$: $\pi = 0.1$ (Weaker)



Weak Instruments: Visualized

Monte Carlo: $Y_i = 0 \cdot D_i + \varepsilon_i$, $D_i = \pi Z_i + \eta_i$: $\pi = 0.01$ (Very Weak)



Many IVs

A thornier problem is many-weak bias, when overidentified

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- But we can have *overfitting* with lots of instruments, which essentially recreates the (endogenous) variation in D_i

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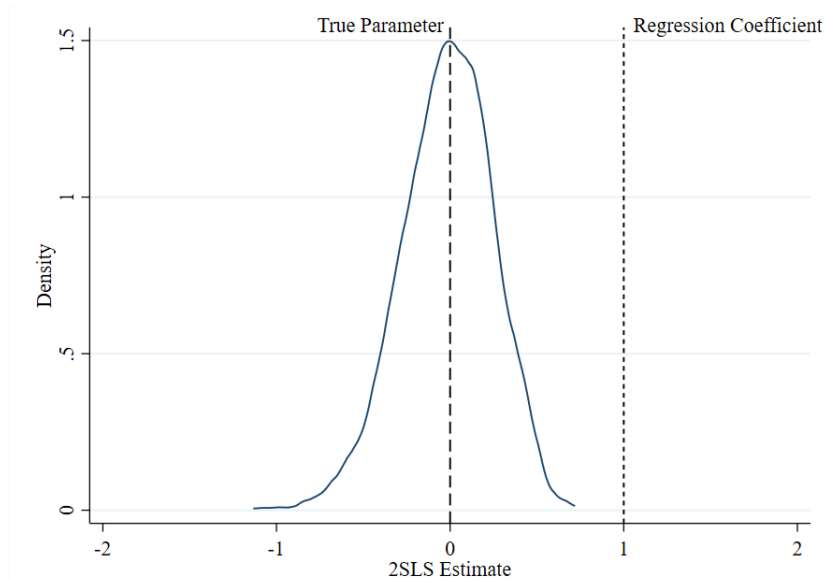
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This became a high-profile problem with Angrist-Krueger '91, where the QOB instrument was interacted with many state/year FEs

- These days folks don't make this mistake ... but many-IV bias can be lurking in other settings with constructed instruments (e.g. judge IV)

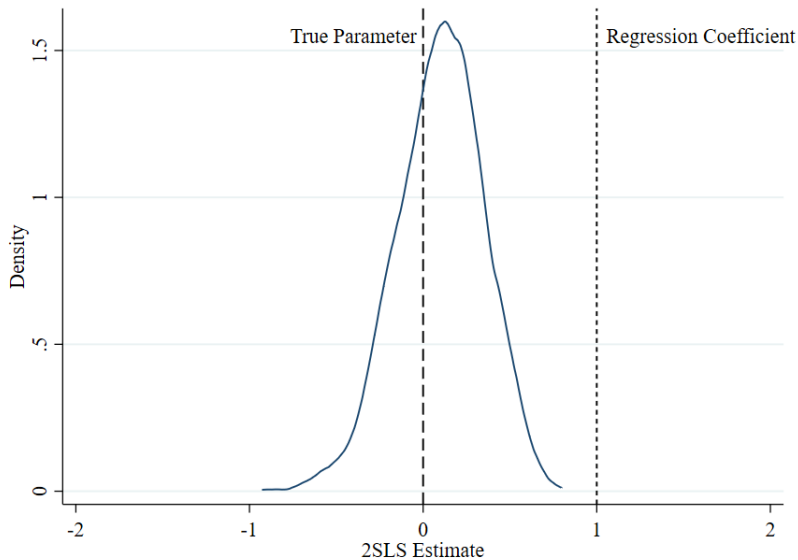
Many Instruments: Visualized

$$Y_i = 0 \cdot D_i + \varepsilon_i, D_i = \pi Z_{i1} + \sum_{\ell > 1} 0 \cdot Z_{i\ell} + \eta_i: \text{IV w/ } Z_{i1} \text{ only}$$



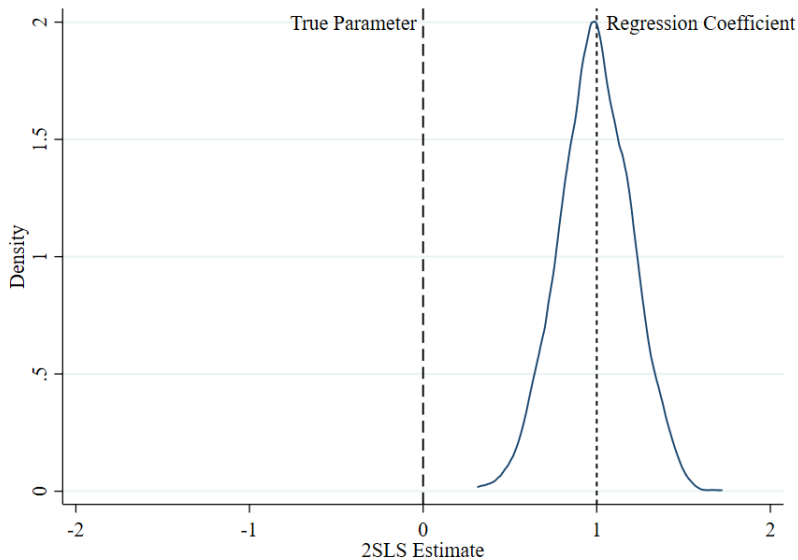
Many Instruments: Visualized

$$Y_i = \alpha \cdot D_i + \varepsilon_i, D_i = \pi Z_{i1} + \sum_{\ell > 1} \pi_{\ell} \cdot Z_{i\ell} + \eta_i \mid W_i / Z_{i1}, \dots, Z_{i10}$$



Many Instruments: Visualized

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What to Do?

Aim for few instruments, and check your F 's after every *ivreg*

- State of the art: Montiel Olea and Pflueger '15; `weakivtest` in Stata
- Staiger-Stock rule-of-thumb ($F > 10$) still seems widely held
- See Lee et al. (2020) and Keane and Neal (2022) for some discussions of additional subtleties

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If your F is small, some things to consider:

- Is there a better functional form for your instrument?
- Do interactions with covariates help? (note: beware many-weak!)
- Does changing the covariate set help? (note: beware invalidity!)
- Check results w/a more robust approach (e.g. Anderson-Rubin, JIVE)

Roadmap

Introductions

IV Mechanics

- Just-Identified IV

- Overidentification

- Weak vs. Many-Weak Bias

Chinese Economic Applications: Session I

Conclusions

IV Methods in Chinese Economic Research

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- Rapid trade liberalization → natural experiments in import competition
- Massive infrastructure investment → exogenous variation in connectivity
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In this section, we examine two landmark papers using IV to study Chinese economic phenomena published in **top economics journals**:

1. **The China Syndrome** (Autor, Dorn, Hanson 2013, *AER*)

Shift-share IV to study effects of Chinese import competition on US labor markets

Paper 1: “The China Syndrome”

Citation: Autor, D., Dorn, D., and Hanson, G.H. (2013). “The China Syndrome: Local Labor Market Effects of Import Competition in the United States.” *American Economic Review*, 103(6): 2121–2168.

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Context: China’s WTO accession (2001) accelerated its integration into world trade. Chinese imports to the US grew from \$100B in 2000 to over \$400B by 2007.

Central Question: Does rising import competition from China cause adverse labor market outcomes in US commuting zones?

- Challenge: Chinese imports may respond to demand conditions in the US that independently affect local employment — *reverse causality and omitted variable bias*

ADH 2013: IV Strategy (Shift-Share)

The Endogeneity Problem:

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IV Solution – Shift-Share Instrument:

$$Z_i = \sum_k \underbrace{s_{ik}}_{\text{share}} \times \underbrace{g_k^{OTH}}_{\text{shock}}$$

- **Shares** s_{ik} : initial (1990) employment share of industry k in CZ i
- **Shocks** g_k^{OTH} : growth of Chinese imports in 8 *other* high-income countries (Australia, Denmark, Finland, Germany, Japan, New Zealand, Spain, Switzerland)

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Data sources:

- US Decennial Census and American Community Survey (employment/wages)
- UN Comtrade (bilateral import flows by industry)
- County Business Patterns (establishment-level employment)

ADH 2013: Original Results

Table 4: Shift-Share IV Estimates of the Effect of Chinese Imports on Manufacturing Employment

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Coefficient	-0.596 (0.114)	-0.489 (0.100)	-0.267 (0.099)	-0.314 (0.107)	-0.310 (0.134)	-0.290 (0.129)	-0.432 (0.205)
<u>Regional controls</u>							
Autor et al. (2013) controls	✓	✓	✓		✓	✓	✓
Start-of-period mfg. share	✓						
Lagged mfg. share		✓	✓	✓	✓	✓	✓
Period-specific lagged mfg. share			✓	✓	✓	✓	✓
Lagged 10-sector shares					✓		✓
Local Acemoglu et al. (2016) controls						✓	
Lagged industry shares							✓
SSIV first stage F -stat.	185.6	166.7	123.6	272.4	64.6	63.3	27.6
# of region-periods	1,444	1,444	1,444	1,444	1,444	1,444	1,444
# of industry-periods	796	794	794	794	794	794	794

ADH 2013: Refined Results (Borusyak, Hull, Jaravel 2022)

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- Columns 3–7 include the key $\sum_k s_{ik} \times (\text{period FE})$ control
- Standard errors / first-stage F-stats computed via *ssaggregate*

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- \$1,000 increase in Chinese import exposure per worker $\rightarrow \approx 2-2.5\%$ decline in the manufacturing employment share in commuting zones

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Why it matters for IV methods:

- Introduced and popularized the **shift-share IV** design in economics
- Borusyak, Hull, and Jaravel (2022) provide formal conditions for validity: key is exogeneity of shocks (not shares)

Paper 2: “On the Road”

Citation: Banerjee, A., Duflo, E., and Qian, N. (2020). “On the Road: Access to Transportation Infrastructure and Economic Growth in China.” *Journal of Political Economy*, 128(8): 2967–3001.

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Context: China built one of the world’s most extensive road and railroad networks during the 1990s–2000s. By 2010, China had the world’s second-largest highway network.

Central Question: Does access to transportation infrastructure (national highways and railroads) cause economic growth in Chinese counties?

- Challenge: roads are built to serve economically active areas → strong *reverse causality*: wealthier counties attract more infrastructure

BDQ 2020: IV Strategy

Identification challenge: Infrastructure placement is endogenous — roads go where economic activity is or is expected to be.

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Instrument construction:

1. Identify major city pairs linked in the 1962 railroad plan
2. Draw straight lines between them — counties near these lines are

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Specification: Panel IV regression with county and year fixed effects,

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- Proximity to a national highway increases GDP growth significantly
- Being within 10km of a highway → roughly 10–15% higher annual GDP growth during the construction period
- Effects are concentrated in manufacturing and trade-related sectors
- Counties with initially weaker connectivity benefit more from access

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- Being within 10km of a highway → roughly 10–15% higher annual GDP growth during the construction period
- Effects are concentrated in manufacturing and trade-related sectors
- Counties with initially weaker connectivity benefit more from access

Why it matters for IV methods:

- Illustrates the “**historical plan as instrument**” strategy — a common approach in development economics
- Distinguishes between actual placement (endogenous) and historically planned routes (exogenous to current economic conditions)

BDQ 2020: Key Results and Lessons

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Validity checks:

- Placebo test: straight-line connections between *non-major* cities do

Comparing the Two Papers

	ADH (2013)	BDQ (2020)
Journal	AER	JPE
Setting	US CZs / China trade	Chinese counties
Treatment	Chinese import competition	Road/rail access
IV type	Shift-share	Historical plan
Exog. source	Imports in other countries	1962 military plan
Unit	CZ $\times$ period	County $\times$ year
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Common thread: Both papers use a “**double-level**” IV logic: exogenous variation exists at a higher/different level (industries/planned routes), which is “translated” down to the unit of interest via a known formula

Roadmap

Introductions

IV Mechanics

- Just-Identified IV

- Overidentification

- Weak vs. Many-Weak Bias

Chinese Economic Applications: Session I

Conclusions

Session 1 Summary

IV Mechanics:

- IV identification: $\beta^{IV} = \rho/\pi$ (reduced form over first stage)
- 2SLS with multiple instruments: efficiency-weighted combination
- Overidentification tests: valid under constant effects, interpret carefully
- Weak and many-IV bias: aim for $F > 10$ in first stage

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Chinese Economics Applications:

- ADH (2013): Shift-share IV reveals large labor market costs of China trade shock
- BDQ (2020): Historical plan IV reveals positive infrastructure effects in China

Preview: Session 2

In Session 2, we will cover:

1. **IV Interpretation**

- What does IV identify? The Local Average Treatment Effect (LATE)
- Imbens-Angrist (1994): monotonicity and complier analysis
- Diff-in-diff and IV: connections and extensions

2. **Formula Instruments**

- Shift-share IV: formal conditions and implementation
- Recentered IV: beyond shift-share (Borusyak-Hull 2023)
- China's High-Speed Rail as a running illustration

3. **Chinese Applications: Session II**

- Meng, Qian, Yared (2015, ReStud): Causes of China's Great Famine

Keep Calm and ivreg2 On!

Thank you for your attention!

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Session 2 continues after a short break.